

# Read Me

The code folder contains 2 “main” files that include the codes to generate the main results for the paper.

The full “main.R” include generating results for both Tree methods and the baseline triple sorting based portfolios.

The “main\_simplified.R” file runs faster and only contains the main results such as Sharpe ratios of the generated “SDF” portfolio and the corresponding asset pricing test results.

The Data folder contains sample intermediate data from the outputs of “step 1” below, which includes yearly data with returns and characteristics (additional noise added to return and raw size numbers to avoid violating WRDS data sharing agreements, see below for more details).

## Data Generation Steps:

**Step 1:** requires the raw input files of tables of characteristics ( $T * N$ ). In this step we combine the returns, size and information about three firm characteristics (which are also transformed into quantiles), saved as yearly files

### Function:

`create_yearly_chunks(y_min,y_max,feats_list,feat1,feat2,input_path,output_path,add_noise=FALSE)`

Note that at this step we also add random noise to the return and market cap data (used for creating value-weighted portfolios), using the following lines below:

```
if (add_noise){  
  n = nrow(data_train)  
  data_train[, 'ret'] = pmax(data_train[, 'ret'] + rnorm(n, 0, 0.05), -0.99)  
  data_train[, 'size'] = pmax(data_train[, 'size'] + rnorm(n, 0, 0.05), 0.01)  
}
```

Input:

| Date     | AC.10001 | AC.10005 | AC.10006   | AC.10007 | AC.10008 | AC.10010 | AC.10011 | AC.10012 | AC.10013 | AC.10014   |
|----------|----------|----------|------------|----------|----------|----------|----------|----------|----------|------------|
| 19630131 | NA       | NA       | 0.05727206 | NA       | NA       | NA       | NA       | NA       | NA       | -0.0695088 |
| 19630228 | NA       | NA       | 0.05727206 | NA       | NA       | NA       | NA       | NA       | NA       | -0.0695088 |
| 19630329 | NA       | NA       | 0.05727206 | NA       | NA       | NA       | NA       | NA       | NA       | -0.0695088 |
| 19630430 | NA       | NA       | 0.05727206 | NA       | NA       | NA       | NA       | NA       | NA       | -0.0695088 |
| 19630531 | NA       | NA       | 0.05727206 | NA       | NA       | NA       | NA       | NA       | NA       | -0.0695088 |
| 19630628 | NA       | NA       | 0.05727206 | NA       | NA       | NA       | NA       | NA       | NA       | -0.0695088 |
| 19630731 | NA       | NA       | 0.08806884 | NA       | NA       | NA       | NA       | NA       | NA       | -0.2045134 |
| 19630830 | NA       | NA       | 0.08806884 | NA       | NA       | NA       | NA       | NA       | NA       | -0.2045134 |
| 19630930 | NA       | NA       | 0.08806884 | NA       | NA       | NA       | NA       | NA       | NA       | -0.2045134 |
| 19631031 | NA       | NA       | 0.08806884 | NA       | NA       | NA       | NA       | NA       | NA       | -0.2045134 |
| 19631129 | NA       | NA       | 0.08806884 | NA       | NA       | NA       | NA       | NA       | NA       | -0.2045134 |
| 19631231 | NA       | NA       | 0.08806884 | NA       | NA       | NA       | NA       | NA       | NA       | -0.2045134 |
| 19640131 | NA       | NA       | 0.08806884 | NA       | NA       | NA       | NA       | NA       | NA       | -0.2045134 |
| 19640228 | NA       | NA       | 0.08806884 | NA       | NA       | NA       | NA       | NA       | NA       | -0.2045134 |
| 19640331 | NA       | NA       | 0.08806884 | NA       | NA       | NA       | NA       | NA       | NA       | -0.2045134 |
| 19640430 | NA       | NA       | 0.08806884 | NA       | NA       | NA       | NA       | NA       | NA       | -0.2045134 |
| 19640528 | NA       | NA       | 0.08806884 | NA       | NA       | NA       | NA       | NA       | NA       | -0.2045134 |
| 19640630 | NA       | NA       | 0.08806884 | NA       | NA       | NA       | NA       | NA       | NA       | -0.2045134 |
| 19640731 | NA       | NA       | -0.1155968 | NA       | NA       | NA       | NA       | NA       | NA       | 0.64516129 |
| 19640831 | NA       | NA       | -0.1155968 | NA       | NA       | NA       | NA       | NA       | NA       | 0.64516129 |
| 19640930 | NA       | NA       | -0.1155968 | NA       | NA       | NA       | NA       | NA       | NA       | 0.64516129 |
| 19641030 | NA       | NA       | -0.1155968 | NA       | NA       | NA       | NA       | NA       | NA       | 0.64516129 |

Sample output:

| yy   | mm | date     | permno | ret       | LME        | BEME       | AC         | size       |
|------|----|----------|--------|-----------|------------|------------|------------|------------|
| 1963 | 1  | 19630131 | 10006  | 0.047002  | 0.77663734 | 0.89846743 | 0.7587007  | 112.062625 |
| 1963 | 1  | 19630131 | 10014  | 0.034483  | 0.35190616 | 0.72988506 | 0.14153132 | 11.94075   |
| 1963 | 1  | 19630131 | 10102  | 0.041475  | 0.87487781 | 0.51149425 | 0.20649652 | 271.95525  |
| 1963 | 1  | 19630131 | 10153  | 0.162393  | 0.79618768 | 0.95402299 | 0.16705336 | 130.9815   |
| 1963 | 1  | 19630131 | 10161  | 0.125     | 0.92082111 | 0.2816092  | 0.774942   | 443.145    |
| 1963 | 1  | 19630131 | 10188  | 0.157407  | 0.50782014 | 0.75478927 | 0.98607889 | 25.2045    |
| 1963 | 1  | 19630131 | 10233  | 0.066176  | 0.81476051 | 0.41187739 | 0.15313225 | 151.504    |
| 1963 | 1  | 19630131 | 10241  | 0.032787  | 0.95845552 | 0.60344828 | 0.15777262 | 747.32625  |
| 1963 | 1  | 19630131 | 10268  | 0.086735  | 0.30840665 | 0.81800766 | 0.15081206 | 9.7265     |
| 1963 | 1  | 19630131 | 10321  | 0.238462  | 0.88220919 | 0.63601533 | 0.60788863 | 300.12125  |
| 1963 | 1  | 19630131 | 10364  | 0.092105  | 0.88807429 | 0.78735632 | 0.36194896 | 311.448    |
| 1963 | 1  | 19630131 | 10372  | 0.098039  | 0.80742913 | 0.91187739 | 0.28074246 | 143.70525  |
| 1963 | 1  | 19630131 | 10401  | 0.036403  | 1          | 0.32950192 | 0.34338747 | 28382.0418 |
| 1963 | 1  | 19630131 | 10460  | 0.021277  | 0.61290323 | 0.31226054 | 0.61484919 | 42.065     |
| 1963 | 1  | 19630131 | 10479  | 0.171875  | 0.7111437  | 0.94636015 | 0.51276102 | 72         |
| 1963 | 1  | 19630131 | 10487  | 0.053719  | 0.73558162 | 0.70306513 | 0.25522042 | 84.54875   |
| 1963 | 1  | 19630131 | 10495  | 0.148148  | 0.91837732 | 0.9559387  | 0.46171694 | 433.9575   |
| 1963 | 1  | 19630131 | 10516  | -0.035928 | 0.69892473 | 0.92911877 | 0.26914153 | 66.34075   |
| 1963 | 1  | 19630131 | 10559  | 0.030387  | 0.82942326 | 0.37931034 | 0.47099768 | 175.389    |
| 1963 | 1  | 19630131 | 10604  | 0.108527  | 0.92130987 | 0.8256705  | 0.68213457 | 444.324375 |
| 1963 | 1  | 19630131 | 10727  | 0.097059  | 0.52883675 | 0.54022989 | 0.91183295 | 28.9425    |
| 1963 | 1  | 19630131 | 10743  | -0.037037 | 0.37829912 | 0.88314176 | 0.99071926 | 13.932     |
| 1963 | 1  | 19630131 | 10751  | 0.041096  | 0.88123167 | 0.62260536 | 0.83990719 | 296.90925  |
| 1963 | 1  | 19630131 | 10786  | 0.091703  | 0.98387097 | 0.70114943 | 0.55684455 | 1302.75238 |
| 1963 | 1  | 19630131 | 10866  | 0.069767  | 0.82355816 | 0.38888889 | 0.6450116  | 162.454    |
| 1963 | 1  | 19630131 | 10874  | 0.040268  | 0.89247312 | 0.06321839 | 1          | 330.29575  |
| 1963 | 1  | 19630131 | 10890  | 0.062222  | 0.83870968 | 0.33716475 | 0.03712297 | 187.846875 |
| 1963 | 1  | 19630131 | 10970  | 0.008571  | 0.96627566 | 0.22222222 | 0.98143852 | 848.44375  |
| 1963 | 1  | 19630131 | 10989  | 0.02457   | 0.97751711 | 0.07854406 | 0.67981439 | 1128.66188 |
| 1963 | 1  | 19630131 | 11017  | 0.157895  | 0.35581623 | 0.12260536 | 0.76102088 | 12.13625   |
| 1963 | 1  | 19630131 | 11084  | 0.083333  | 0.15249267 | 0.6532567  | 0.90023202 | 4.3065     |
| 1963 | 1  | 19630131 | 11092  | 0.254902  | 0.70381232 | 0.95210728 | 0.50812065 | 68.42925   |
| 1963 | 1  | 19630131 | 11105  | 0.198413  | 0.64418377 | 0.16475096 | 0.5800464  | 49.59675   |
| 1963 | 1  | 19630131 | 11164  | 0.032258  | 0.77956989 | 0.52873563 | 0.61020882 | 115.362625 |

**Step 2:** We then use the output from the previous step to generate tree portfolios. We need value-weighted returns at all levels of the trees, and the characteristics min/max (for plotting portfolio boundary in heatmaps and other applications).

When tested on a MacBook Pro (2020, M1 CPU, 8 cores, 16GB RAM), it takes about 1 hour to run the code with single-core processing, and about 20 mins to do it in parallel, using 6 cores (that is, using paralleln=6)

### Function:

create\_tree\_portfolio(y\_min,y\_max,tree\_depth,feats\_list,feat1,feat2,input\_path,output\_path,r  
unparallel,pralleln)

### Output:

Value-weighted returns for a single tree:

|            | 1          | 11         | 12         | 111        | 112        | 121        | 122        | 1111       | 1112       | 1121       | 1122       | 1211       | 1212       | 1221       | 1222 |
|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------|
| 0.02820257 | 0.02067127 | 0.03105146 | 0.02895766 | 0.01683286 | 0.03152647 | 0.03063924 | 0.04949735 | 0.01823685 | 0.01450051 | 0.01856592 | 0.04882498 | 0.0207837  | 0.03500267 | 0.02091045 |      |
| 0.01733193 | 0.02035432 | 0.01624687 | 0.02116493 | 0.01994702 | 0.01496484 | 0.01729267 | 0.0298231  | 0.01765641 | 0.02537512 | 0.01455486 | 0.01059904 | 0.0180842  | 0.01734049 | 0.01717033 |      |
| 0.01710811 | 0.03804182 | 0.01000118 | 0.05046387 | 0.03004293 | 0.01455366 | 0.00583811 | 0.07292696 | 0.04275164 | 0.034988   | 0.02271261 | 0.03611581 | 0.00654568 | 0.00398395 | 0.00899957 |      |
| 0.00554728 | -0.0161195 | 0.01290732 | -0.0218917 | -0.0131336 | 0.01381794 | 0.01230508 | -0.0390815 | -0.0149058 | -0.0224763 | 0.00052066 | 0.00172565 | 0.02388299 | 0.01187657 | 0.01322847 |      |
| 0.01570791 | 0.00553586 | 0.01930198 | 0.01211934 | 0.00261677 | 0.02301361 | 0.01696538 | 0.00776625 | 0.01446815 | 0.00386218 | 0.00124667 | 0.02509091 | 0.02179635 | 0.01724343 | 0.01636117 |      |
| 0.01691577 | 0.01385566 | 0.01794572 | 0.00962256 | 0.0163978  | 0.00900567 | 0.02560444 | 0.01852914 | 0.0060022  | 0.0175756  | 0.01562468 | 0.00551077 | 0.01092942 | 0.02851551 | 0.01919816 |      |
| 0.0206998  | 0.01990832 | 0.02098098 | 0.01119789 | 0.02500286 | 0.01940239 | 0.02226742 | 0.01606675 | 0.00831433 | 0.01972444 | 0.02765156 | 0.02776647 | 0.01514681 | 0.0218806  | 0.02293015 |      |
| -0.0110338 | -0.0136683 | -0.0100403 | -0.0114647 | -0.0148909 | -0.0109337 | -0.0090218 | -0.013675  | -0.0100716 | -0.0099482 | -0.0170918 | -0.0110594 | -0.0108817 | -0.0105112 | -0.0068767 |      |
| 0.02976584 | 0.05038109 | 0.02265543 | 0.0587096  | 0.04584216 | 0.02772902 | 0.01669803 | 0.05495292 | 0.06094994 | 0.04285702 | 0.04738379 | 0.03326184 | 0.02543846 | 0.00914941 | 0.02759731 |      |
| 0.00996298 | 0.00583474 | 0.01136425 | -0.0008531 | 0.0093084  | 0.01279309 | 0.00965951 | -0.0218361 | 0.01255418 | 0.00115601 | 0.01746025 | 0.02579383 | 0.00683982 | 0.00256229 | 0.01939438 |      |
| 0.00231134 | 0.00020818 | 0.00307172 | -0.0151216 | 0.00833709 | 0.00847811 | 0.00047947 | -0.0485933 | 0.00202495 | -0.0107068 | 0.01807859 | 0.0012482  | 0.01480476 | -0.001415  | 0.00635108 |      |
| 0.00371864 | -0.0068914 | 0.00736597 | -0.0158503 | -0.0019324 | -0.004009  | 0.01300891 | -0.0132955 | -0.0171634 | -0.0045719 | 0.00040015 | -0.002461  | -0.0056915 | 0.00587252 | 0.02426515 |      |
| 0.03693159 | 0.06247113 | 0.0284583  | 0.05889926 | 0.0640598  | 0.03944428 | 0.02272311 | 0.04811814 | 0.06456235 | 0.05214918 | 0.07375184 | 0.04134053 | 0.03774255 | 0.02359305 | 0.02163422 |      |
| 0.00453549 | 0.02553253 | -0.0016825 | 0.04603465 | 0.01551967 | 0.00075444 | 0.0277827  | 0.04475287 | 0.04667083 | 0.01026278 | 0.02124422 | 0.00052808 | 0.00093862 | 0.00238578 | -0.0106763 |      |
| -0.0114695 | -0.0032528 | -0.0139305 | -0.0100045 | 0.00029697 | -0.0107146 | -0.015581  | -0.0079866 | -0.0107878 | -0.0030165 | 0.0054495  | -0.006837  | -0.0136693 | -0.0090209 | -0.0260953 |      |
| 0.03450713 | 0.03459324 | 0.03447783 | 0.0323604  | 0.03569771 | 0.02485824 | 0.04226687 | 0.05840577 | 0.02139975 | 0.04131453 | 0.03048734 | 0.02558781 | 0.02454495 | 0.0480667  | 0.03544746 |      |
| -0.0042536 | -0.007497  | -0.0029669 | -0.014265  | -0.0031793 | 0.00170501 | -0.0049132 | -0.0033844 | -0.0181787 | -0.0042791 | -0.0024535 | -0.0022891 | 0.00402664 | 0.00459684 | -0.0135495 |      |
| -0.0483298 | -0.0611921 | -0.0428863 | -0.0698317 | -0.0550803 | -0.046469  | -0.0399286 | -0.0825816 | -0.0653728 | -0.0582868 | -0.052974  | -0.0522494 | -0.0441457 | -0.0432065 | -0.0386622 |      |

Portfolio characteristics min/max for a single tree (there are 2\*3 such files for one single tree)

|            | 1          | 11         | 12         | 111        | 112        | 121        | 122        | 1111       | 1112       | 1121       | 1122       | 1211        | 1212       | 1221       | 1222 |
|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|-------------|------------|------------|------|
| 0.04949054 | 0.04949054 | 0.07763222 | 0.04949054 | 0.12372635 | 0.19505095 | 0.07763222 | 0.04949054 | 0.55458515 | 0.12372635 | 0.7130034  | 0.19505095 | 0.77438137  | 0.07763222 | 0.81077147 |      |
| 0.05009634 | 0.05009634 | 0.07273603 | 0.05009634 | 0.11247592 | 0.20712909 | 0.07273603 | 0.05009634 | 0.54238921 | 0.11247592 | 0.72976879 | 0.20712909 | 0.7605973   | 0.07273603 | 0.82803468 |      |
| 0.07218479 | 0.07218479 | 0.07410972 | 0.07218479 | 0.13811357 | 0.1732435  | 0.07410972 | 0.07218479 | 0.58373436 | 0.13811357 | 0.72040423 | 0.1732435  | 0.77815207  | 0.07410972 | 0.80750722 |      |
| 0.05371703 | 0.05371703 | 0.08201439 | 0.05371703 | 0.1294964  | 0.13285372 | 0.08201439 | 0.05371703 | 0.56834532 | 0.1294964  | 0.72613909 | 0.13285372 | 0.77410072  | 0.08201439 | 0.81870504 |      |
| 0.04432793 | 0.04432793 | 0.10438513 | 0.04432793 | 0.13393708 | 0.10438513 | 0.17254528 | 0.04432793 | 0.56053384 | 0.13393708 | 0.73879886 | 0.10438513 | 0.76120114  | 0.17254528 | 0.82602479 |      |
| 0.03807711 | 0.03807711 | 0.11851499 | 0.03807711 | 0.12946216 | 0.11851499 | 0.17705854 | 0.03807711 | 0.5578296  | 0.12946216 | 0.71727749 | 0.11851499 | 0.78724417  | 0.17705854 | 0.80485483 |      |
| 0.0418251  | 0.0418251  | 0.10931559 | 0.0418251  | 0.12357414 | 0.10931559 | 0.21910646 | 0.0418251  | 0.58555133 | 0.12357414 | 0.72576046 | 0.10931559 | 0.77519011  | 0.21910646 | 0.80750951 |      |
| 0.04356061 | 0.04356061 | 0.10748106 | 0.04356061 | 0.13020833 | 0.10748106 | 0.21448864 | 0.04356061 | 0.58570076 | 0.13020833 | 0.72159091 | 0.10748106 | 0.81013258  | 0.21448864 | 0.79640152 |      |
| 0.05188679 | 0.05188679 | 0.10660377 | 0.05188679 | 0.15707547 | 0.10660377 | 0.1995283  | 0.05188679 | 0.58537736 | 0.15707547 | 0.71886792 | 0.10660377 | 0.81037736  | 0.1995283  | 0.81320755 |      |
| 0.04674221 | 0.04674221 | 0.09915014 | 0.04674221 | 0.13739377 | 0.09915014 | 0.20443815 | 0.04674221 | 0.58829084 | 0.13739377 | 0.73040604 | 0.09915014 | 0.79697828  | 0.20443815 | 0.81539188 |      |
| 0.04194156 | 0.04194156 | 0.10556079 | 0.04194156 | 0.14938737 | 0.10556079 | 0.19509896 | 0.04194156 | 0.58930254 | 0.14938737 | 0.69792648 | 0.10556079 | 0.79971725  | 0.19509896 | 0.81762488 |      |
| 0.04701457 | 0.04701457 | 0.10202163 | 0.04701457 | 0.12717442 | 0.10202163 | 0.18758815 | 0.04701457 | 0.59379408 | 0.12717442 | 0.6906441  | 0.10202163 | 0.79360602  | 0.18758815 | 0.82816173 |      |
| 0.04515527 | 0.04515527 | 0.19567767 | 0.04515527 | 0.1335847  | 0.19807446 | 0.19567767 | 0.04515527 | 0.579497   | 0.1335847  | 0.70178738 | 0.19807446 | 0.806657921 | 0.19567767 | 0.87367127 |      |

**Step 3:** in this step we combine the tree portfolios from all trees into one file and calculate their excess return over risk free rates and remove duplicates of the upper-level tree portfolios.

### Function:

combinetrees(feats\_list, feat1, feat2, tree\_depth, factor\_path, tree\_sort\_path\_base)

Output: value-weighted excess returns for all the nodes of the created, the format is similar to that of characteristics min/max

[illegible]

Empirical results in the paper rely on a larger grid search along the full parameter space:

```
lambda0 = seq(0, 0.9, 0.05)
```

```
lambda2 = 0.1^seq(5, 8, 0.25)
```

In the simplified version of the main code, we use  $\lambda_0=0.15$  and  $\lambda_2=0.1^8$ . When tested on a MacBook Pro (2020, M1 CPU, 8 cores, 16GB RAM), it takes about 1 minute to run the code using 6 cores in parallel.

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